

CAN A PRETRAINED MODEL BEAT TUNED XGBOOST?

I replicated TabPFN v2 (Nature, 2025) on 4 business datasets against 6 gradient-boosting setups. Free Kaggle GPU. No cherry-picking.

4

DATASETS

9

MODELS

900

MODEL FITS

1

FREE T4

Strong default. Not a GBDT killer.

1

WIN

Bank marketing
AUC 0.938 vs 0.928

2

TIES

Credit-G
Telco churn

1

LOSS

Adult income
-0.006 AUC

Scored against the best tree model on each dataset, 25 folds, corrected stats.

No tuning needed: it fits in under a second. That is the real win.

How I tested it

4

DATASETS

credit-g • telco-churn •
bank-marketing • adult

9

MODELS

TabPFN, 3 GBDTs ×
default/tuned, RF, LogReg

25

CV FOLDS EACH

5×5 repeated stratified

15×3

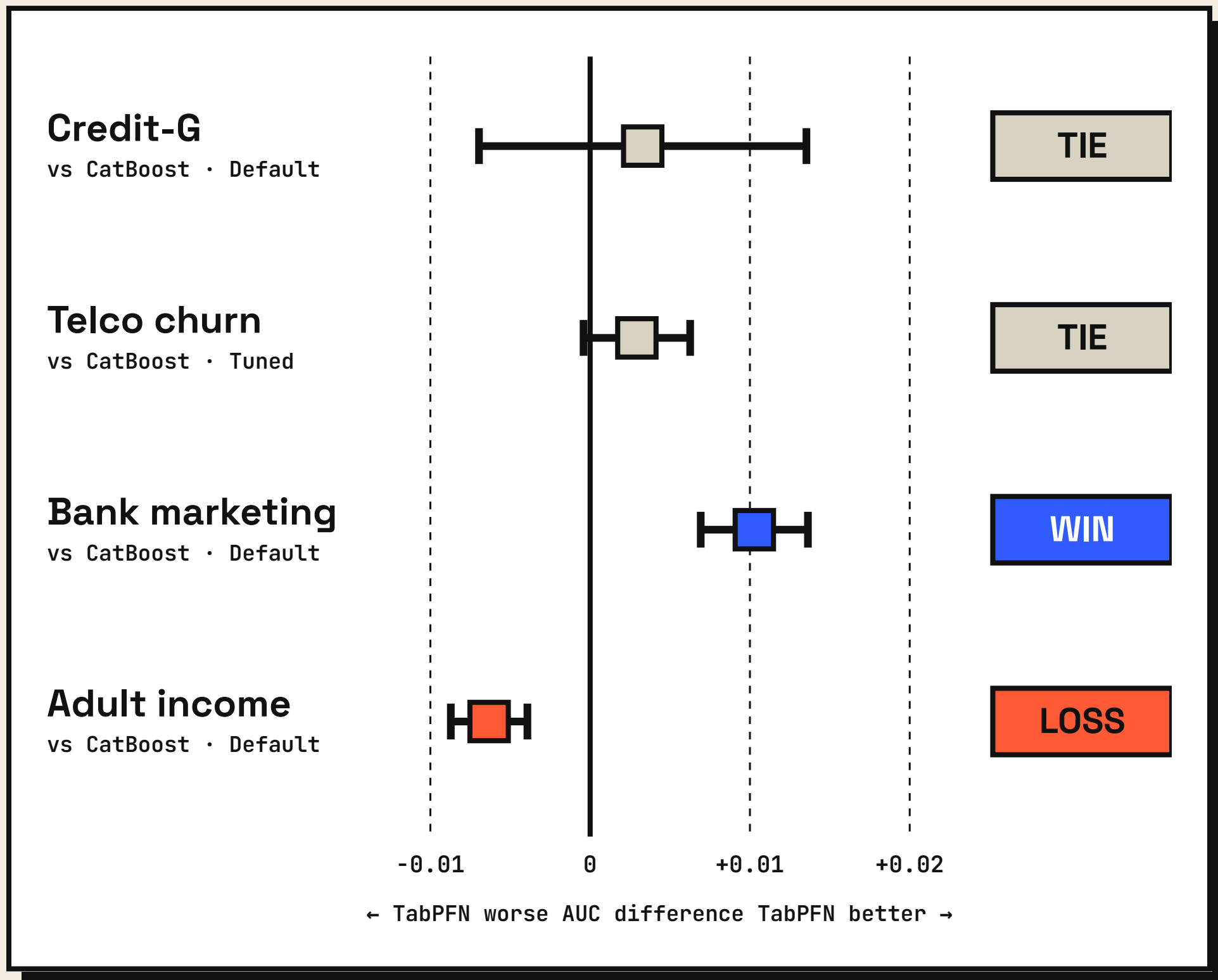
NESTED TUNING

random trials × inner folds

Stats use the Nadeau-Bengio corrected t-test. CV folds overlap, so plain tests overstate confidence.

Free Kaggle T4. Open v2 weights, commercial use OK.

Vs the best baseline



Square = mean AUC gap over 25 folds. Line = corrected 95% interval. Line crossing 0 means a tie.

A naive Wilcoxon test calls Telco a win. The corrected test says tie.

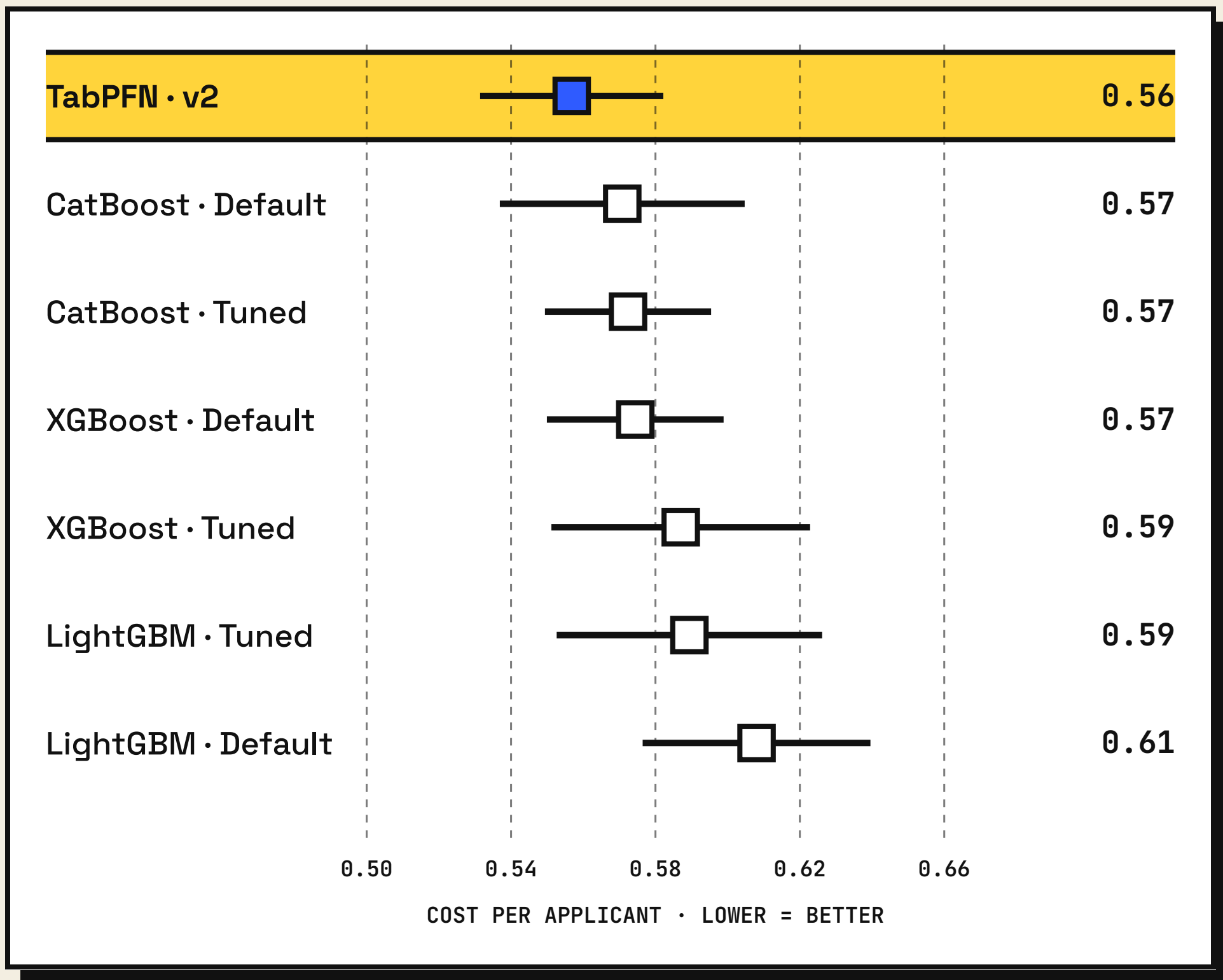
Beats untuned. Ties tuned.

	Credit-G	Telco churn	Bank marketing	Adult income
CatBoost • Default	TIE +0.003	WIN +0.005	WIN +0.010	LOSS -0.006
CatBoost • Tuned	TIE +0.008	TIE +0.003	WIN +0.011	LOSS -0.006
XGBoost • Default	TIE +0.018	WIN +0.027	WIN +0.024	TIE +0.001
XGBoost • Tuned	TIE +0.009	TIE +0.004	WIN +0.018	LOSS -0.006
LightGBM • Default	WIN +0.023	WIN +0.014	WIN +0.016	LOSS -0.004
LightGBM • Tuned	TIE +0.014	WIN +0.003	WIN +0.015	LOSS -0.005

Cell = TabPFN AUC minus that model. WIN or LOSS only when the corrected interval excludes 0. No multiple-comparison correction.

Biggest edge: default XGBoost and LightGBM. Tuning closes the gap.

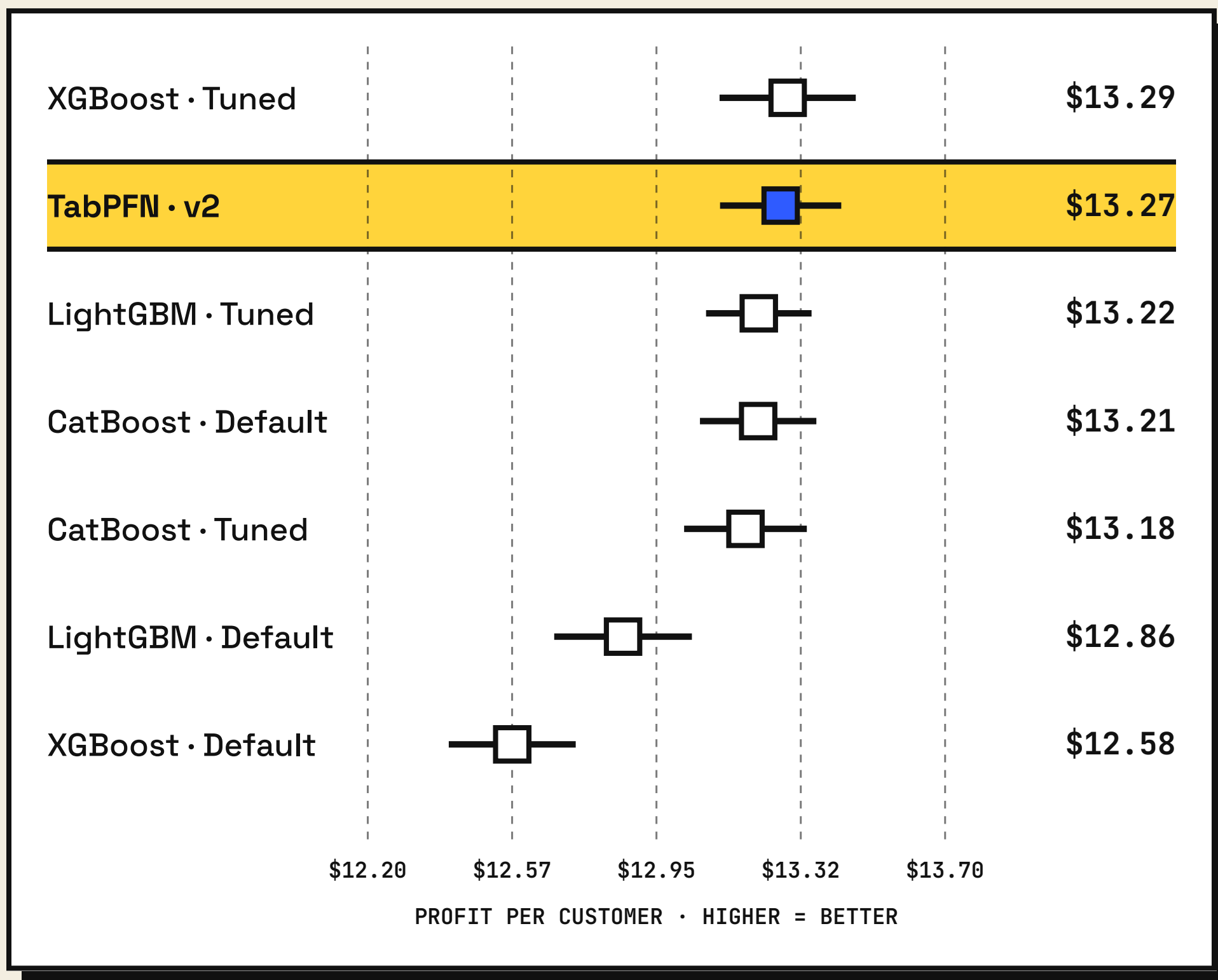
Lowest cost. Within noise.



Approving a bad loan costs 5× rejecting a good one. Naive 95% intervals across folds.

0.557 vs 0.572 for tuned CatBoost. Interval spans zero.

Same story. Cents, not dollars.



My assumptions: \$100 customer value, \$20 offer, contact when score > 0.20.

+\$0.41 to +\$0.70 vs untuned LightGBM / XGBoost. ~\$0 vs tuned.

Fast to fit. Slow to serve.

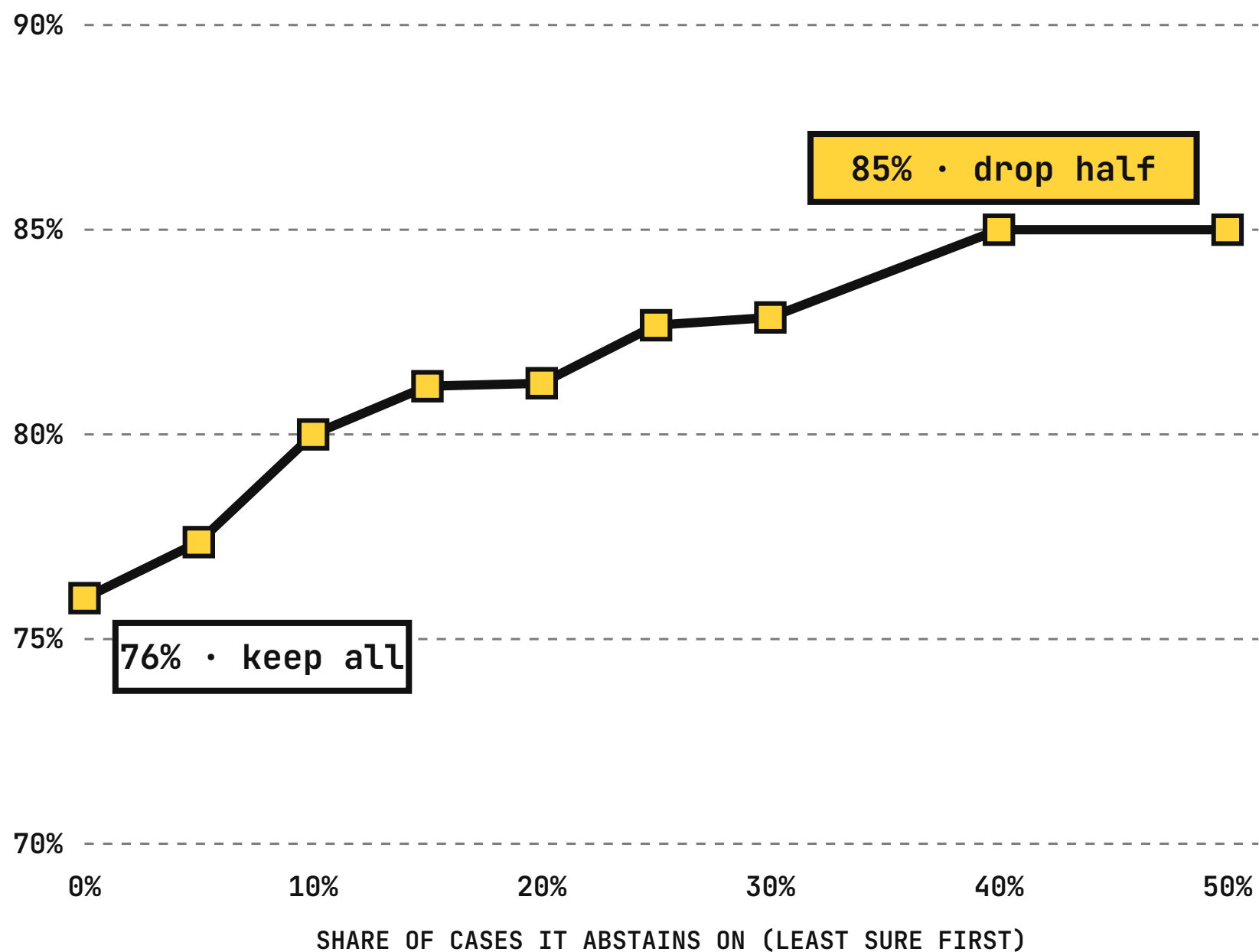


Per-row prediction vs default CatBoost. Fit time: TabPFN 0.3-0.9s, tuned CatBoost 44-68s.

Batch scoring: fine. Real-time at scale: no.

Let it say “I don’t know.”

ACCURACY ON THE CASES IT KEEPS



Caveat: one holdout split of ~200 rows. Noisy. A hint, not a result.

Drop the least-sure half: accuracy 76% → 85%.



Use it. Know its limits.

USE IT WHEN

Under ~10k rows

No time to tune

Batch scoring

Need a strong baseline today

SKIP IT WHEN

Real-time serving

Large data, 500+ features

Tuned GBDT already good

Latency budget is tight

Limits: 4 datasets, 25 folds, no multiple-comparison correction. My planned scaling test crashed, so it is not shown.

Day 02 tomorrow. Follow along. github.com/Shrinet82